

Agenda: Camera Calibration

○ Calibrated Cameras

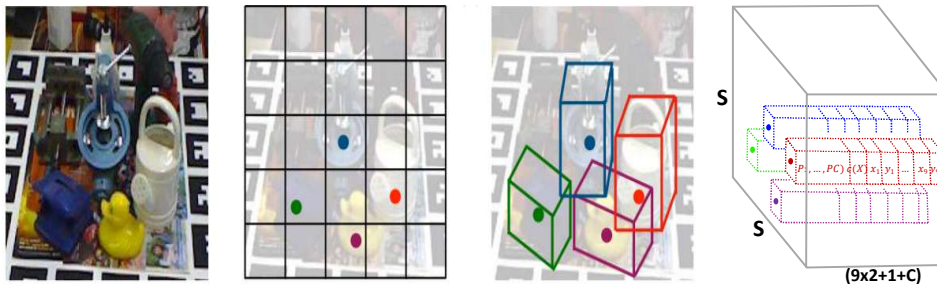
- Why
- **What**
- **How** to calibrate
 - Method 1
 - Method 2
 - Method 3



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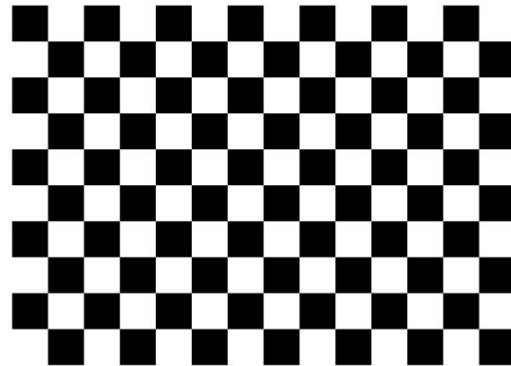
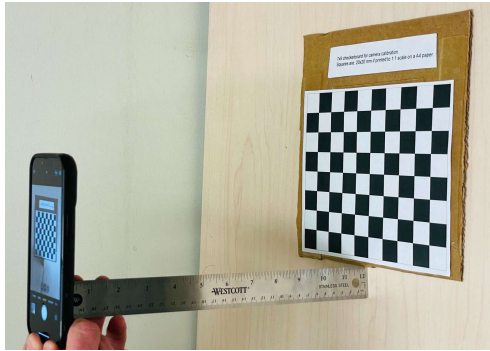
Geometric Camera Calibration: Why

- To get to (e.g.) 6D pose (in camera view) we need to obtain the parameters of the (unknown) camera
 - We need to go from 2D to 3D



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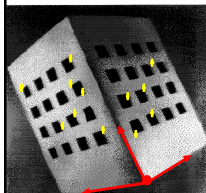
How: Simple Method to get K



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Camera Calibration With Planar Patterns

- The non-planar pattern 'box' may be difficult to carry
- (Multi)Planar patterns are commonly available or can be taped to omnipresent planar objects
 - General method to retrieve camera intrinsics
 - Also uses DLT

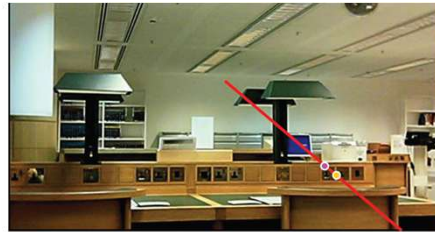


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Other parameters



What property of the world is violated in this image?



Want this image. Cannot use the pinhole camera model.

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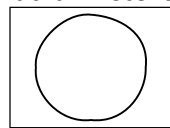
Updated Problem Definition

- Goal:
 - Determine K , camera intrinsics
 - Determine R, T , camera extrinsics
- Refined definition:
 - Also obtain distortion parameters
 - (after R , but before K ; LHS is the actual distorted pixel).



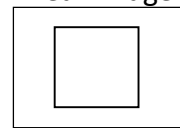
$$\begin{aligned}
 x &= x_d(1 + k_1 r^2 + k_2 r^4) \\
 y &= y_d(1 + k_1 r^2 + k_2 r^4) \\
 r^2 &= x_d^2 + y_d^2
 \end{aligned}$$

Radial Distortion



Correction

Linear Image



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6D Pose via CNN

- To get to (e.g.) 6D pose (in camera view) we need to obtain the parameters of the (unknown) camera
 - We need to go from 2D to 3D

